

(a) Legal repeated-token request

text $\rightarrow$ tokenizer $\rightarrow$ prefix + repeats



$$V = \mathbf{1}v^T$$

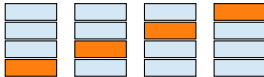
all rows identical

Fresh row permutations leave V unchanged.

Attacker observes protected caches; server secrets remain unavailable.

(b) A small observable orbit

$$C_j = uw^T + s_j a^T$$



$b!$ permutations $\rightarrow b$ marker states

(c) Transfer within a key epoch

state differences $\rightarrow$ equivalent secrets



new request cache $\rightarrow$ row demixing



public dictionary $\rightarrow$ token multiset